



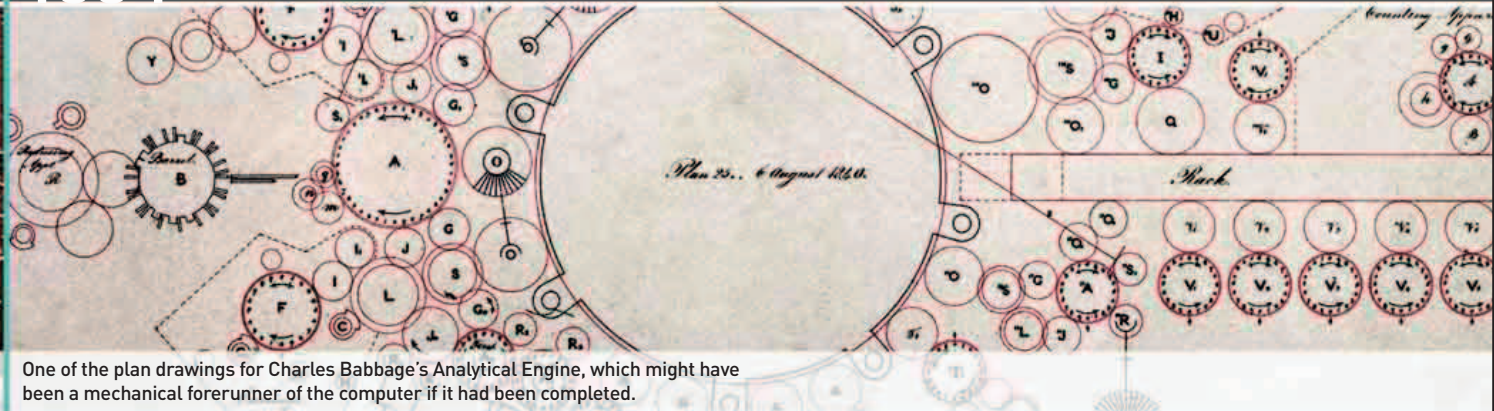
Louis Agassiz's work on fossils gave impetus to the study of extinct life.

BIOCHEMISTRY, THE STUDY OF THE CHEMISTRY OF LIVING THINGS, can be said to have begun in 1833 with the **discovery and isolation of the enzyme diastase** by French chemist **Anselme Payen** (1795–1871). Enzymes are produced by living organisms and act as a catalyst to bring about biochemical reactions (see 1893–94). Diastase is the enzyme in beer mash that helps the starch in barley seed change into soluble sugars.

The term “scientist” was also coined this year by the polymath **William Whewell** (1794–1866). Until this time the only terms in use were “natural philosopher” and “man of science.”

British physician and physiologist **Marshall Hall** (1790–1857) **discovered the reflex arc**—the primitive part of the body's nervous system. It takes time for the brain to receive sense signals, then process and act on them. Reflex arcs provide a rapid automatic response by short-circuiting the brain. When the hand, for example, touches something hot, the sense signal only goes as far as the spinal cord before a message is sent back to move the hand.

German mathematician **Carl Friedrich Gauss** and physicist **Wilhelm Weber** (1804–91) developed the **first practical electromagnetic telegraph**.



One of the plan drawings for Charles Babbage's Analytical Engine, which might have been a mechanical forerunner of the computer if it had been completed.

BRITISH SCIENTIST MICHAEL FARADAY had conducted some brilliant work in 1833 on electrolysis—the chemical reactions that occur when an electric current passes through a liquid (see panel, opposite). In 1834, he published his two laws relating to it. Faraday's **first law of electrolysis** states that the amount of **chemical change varies in proportion to the current**. The second law states that the amount of material deposited on the electrodes by the reaction is proportional to the mass of the material involved in the reactions.

Another law relating to electricity was developed by Russian physicist **Heinrich Lenz** (1804–65). **Lenz's law** stated that an electric current, induced by an electromagnetic field for example, always creates a magnetic force counter (opposite) to the force inducing it.

French engineer **Émile Clapeyron** (1799–1864)

Charles Babbage's Analytical Engine

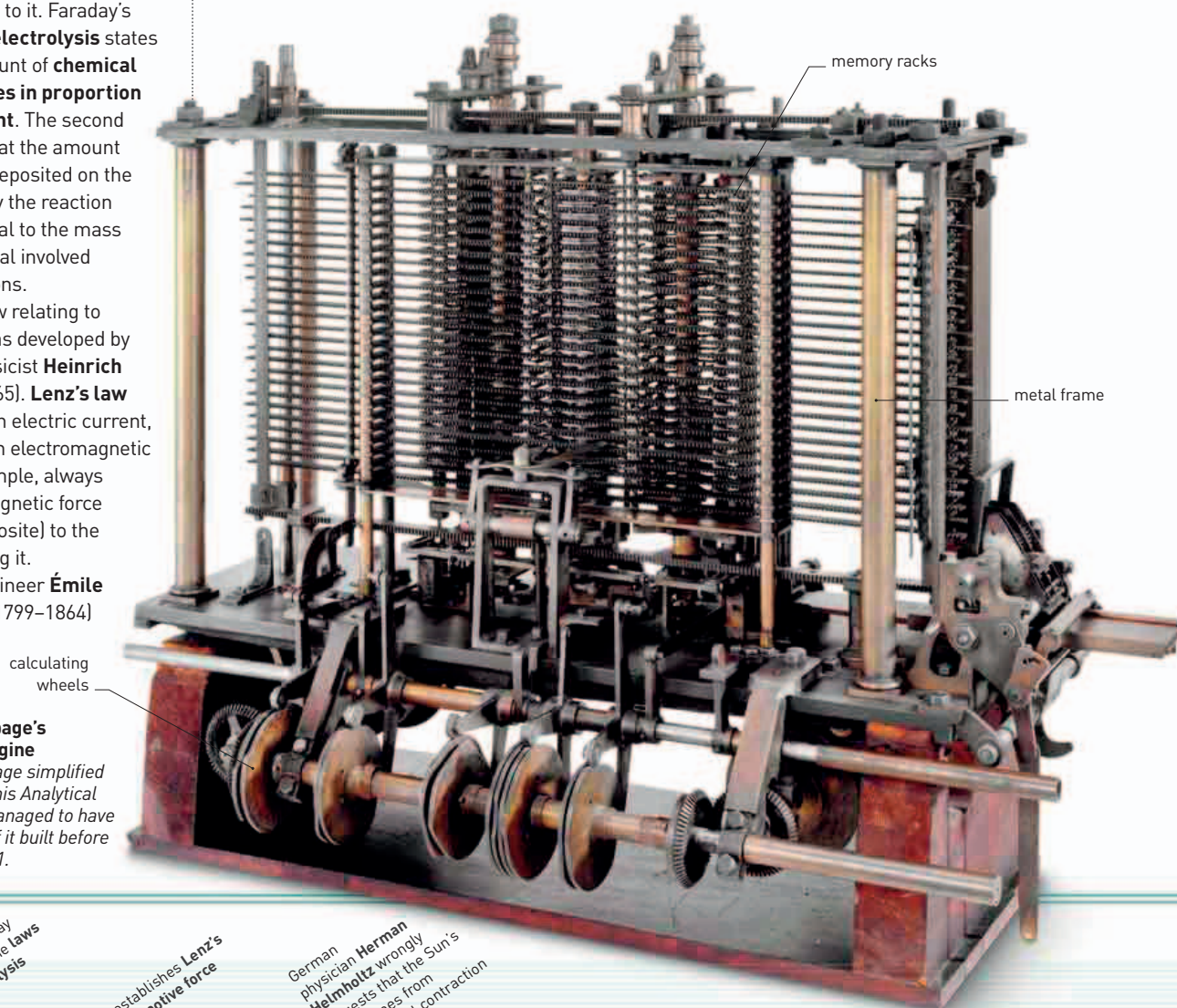
Charles Babbage simplified the design of his Analytical Engine and managed to have a small part of it built before he died in 1871.

began to formulate a third key scientific law. Clapeyron took the work of fellow French physicist Nicolas Léonard Sadi Carnot (see 1823–24) on heat engines and presented it in graphic form. It clarified Carnot's observation that in energy exchanges, the

energy available to make things happen (potential energy) must always gradually run down unless more energy is put in. Burning fuel is an irreversible process, which is why a car needs continual refueling. This became the **basis of the second**

law of thermodynamics (see 1849–51).

Also in 1834, after producing his mechanical calculator, or Difference Engine (see 1822), British inventor Charles Babbage (1791–1871) began to work on plans for an **Analytical Engine**.



calculating wheels

memory racks

metal frame

Anselme Payen discovers the **first enzyme, diastase**, in beer malt

The term “**scientist**” is coined

Marshall Hall discovers the **reflex arc**

Carl Friedrich Gauss and Wilhelm Weber develop the **first electric telegraph**

Michael Faraday publishes the **laws of electrolysis**

Henry Lenz establishes **Lenz's law for electromotive force**

German physician **Herman Helmholtz** wrongly suggests that the Sun's heat comes from gravitational contraction

March 14 British astronomer **John Herschel** discovers the **NGC 3630 star cluster**

June 21 American inventor **Cyrus McCormick** patents his **corn reaper**

Émile Clapeyron presents Nicolas Léonard Sadi Carnot's work on **heat engines**